Lab 1: Introduction to MIPS Assembly

Student Name	Qimin Zhu	Student ID	6888143

Objectives:

- Learn about the MIPS assembly language.
- Write simple MIPS programs and run them in the MARS simulator.
- Use system calls (syscall) for simple input and output.

Outline:

- Check the Code Answer Template — read this first. Each code answer should follow this template.
- Questions Q1.1 – Q1.5: In-Lab Tasks from Section 2.6 of the **MIPS Lab 1** sheet.

How to Complete This Worksheet:

- Complete this worksheet individually and submit your own copy.
- Write and test each answer in **MARS** first.
  - Assemble (F3)
  - Run (F5)
  - Check the output in the Run I/O window.
- Prepare each answer using the **Code Answer Template (page 2)**, then copy-paste your final working code into the grey answer box for each question.
- **Comment every code line** in your own words, very briefly, describing what that line does.
- Your code answers should be **complete and runnable in MARS.**
  - **Grading scores are given to assembling code that runs in MARS.**
  - **I**nclude the header comments,
  - `.data` segments
  - `.text` segments,
  - comments to show your understanding to get a score, if there's a problem assembling your code.

Code Answer Template

Answer each Lab question as a complete MIPS program in the format below.

Steps:

1. Open the **MARS editor tab.**
    a. Fill in the **header information** lines:
        i. Title,
        ii. Author (your name and student ID),
        iii. Date,
        iv. Description
        v. Input / Output of a test run.
    b. Declare your strings and variables in the `.data` segment
    c. Write your instructions in the `.text` segment after `main:`.
    d. **Exit your program** with system call: e.g.
        i. `li $v0, 10`
        ii. `syscall`
2. Make sure your code assembles, runs, and answers the question.
3. **Comment each line of code** with `#`, in your own words, briefly, e.g.
    a. `move $t0, $v0  # Copies $v0's value into register $t0.`
4. **Paste your working code into the answer box.**
    a. We grade according to whether you answered the question, and your code assembles and runs in MARS.

```
# Title:       [Program title, e.g. Echo an Integer]
# Author:      [Your Name] [Your Student ID]
# Date:        [DD/MM/YYYY]
# Description: [One sentence: what does the program do?]
# Input:       [The test input you used, e.g. 20]
# Output:      [The exact output MARS printed, e.g.]
#   Enter an integer: 20
#   You entered: 20
################## Data segment ##################
.data
prompt:  .asciiz "Enter an integer: "         # Your strings / variables here
result:  .asciiz "You entered: "
################## Code segment ##################
.text
.globl main                  # mark the “main” function as globally visible.

main:                        # main function entry
    li   $v0, 4              # Service 4 = print string
    la   $a0, prompt         # Load the address of the prompt text
    syscall                  # Print the prompt
    li   $v0, 5              # Service 5 = read integer
    syscall                  # Read the integer into $v0
    move $t0, $v0            # Copy the input value into $t0
    li   $v0, 4              # Service 4 = print string
    la   $a0, result         # Load the address of the result text
    syscall                  # Print the result text
    li   $v0, 1              # Service 1 = print integer
    move $a0, $t0            # Put the input value in $a0
    syscall                  # Print the value
    # Your code may continue here ...
    # Always end your program with the exit system call ...
```

Part 1 - Lab Tasks (Lab Sheet Section 2.6)

Answer every question using the *Code Answer Template*.

Each answer box should contain one complete, runnable, commented MIPS program.

Question Q1.1

Question / Instruction

Q1.1 Modify the program shown in Figure 2.4 of the lab sheet.

Ask the user to enter an integer value, and then print the result of **doubling** that number.
Use the **add** instruction.
Comment your code, complete the code template format given above.

```
Paste your complete MIPS code answer here
# Title:       double the input
# Author:      Qimin Zhu 6888143
# Date:        02/09/2026
# Description: It will double the value of your input
# Input:       2
# Output:      4
#   Enter an integer: 2
#   After calculation: 4
################### Data segment ###################
.data
prompt:  .asciiz "Enter an integer: "        # Your strings / variables here
result:  .asciiz "After calculation: "
################### Code segment ###################
.text
.globl main                 # mark the “main” function as globally visible.

main:                       # main function entry
   li   $v0, 4              # Service 4 = print string
   la   $a0, prompt         # Load the address of the prompt text
   syscall                  # Print the prompt
   li   $v0, 5              # Service 5 = read integer
   syscall                  # Read the integer into $v0
   move $t0, $v0            # Copy the input value into $t0
   add $t0, $t0, $t0        # double the value
   li   $v0, 4              # Service 4 = print string
   la   $a0, result         # Load the address of the result text
   syscall                  # Print the result text
   li   $v0, 1              # Service 1 = print integer
   move $a0, $t0            # Put the input value in $a0
   syscall                  # Print the value
```

Question Q1.2

Question / Instruction

Q1.2
Ask the user whether they want to repeat the program: `"\nRepeat [y/n]? "`.
Use service code **12** to read a character, and a **branch** instruction to repeat the `main` function if the user inputs the character `'y'`.

Comment your code, complete the code template format given.

Paste your complete MIPS code answer here

```
 # Title:       repeat a loop

# Author:      Qimin Zhu 6888143

# Date:        02/09/2026

# Description: It will double the value of your input after you enter y

# Input:       2

# Output:      4

#   Enter an integer: 2

#   2 now become to: 4

################## Data segment ##################

.data

prompt:  .asciiz "Repeat[y/n]?: "         # Your strings / variables here

result:  .asciiz "You entered: "

################## Code segment ##################

.text

.globl main                  # mark the “main” function as globally visible.


main:                        # main function entry

    li $t1, 'y'

    li $t2, 2

loop:

    li   $v0, 4              # Service 4 = print string

    la   $a0, prompt         # Load the address of the prompt text

    syscall                  # Print the prompt

    ##################

    li   $v0, 12              # Service 12 = read char

    syscall                  # Read the integer into $v0

    move $t0, $v0            # Copy the input value into $t0

    ##################

    beq $t0, $t1, isRepeat

    j exit_program

isRepeat:

    add $t2, $t2, $t2

    li   $v0, 4              # Service 4 = print string
```

```
    la   $a0, result         # Load the address of the result text
    syscall                  # Print the result text
    li   $v0, 1              # Service 1 = print integer
    move $a0, $t2            # Put the input value in $a0
    syscall                  # Print the value
    li $v0, 11
    li $a0, '\n'
    syscall
    j loop
exit_program:
    li   $v0, 11
    li   $a0, '\n'
    syscall
    li   $v0, 1
    move   $a0, $t2
    syscall
    li $v0, 10
    syscall
```

Question Q1.3

Question / Instruction

Q1.3
Write a MIPS program that asks the user to input their name, and then prints "Hello ", followed by the name entered by the user.

Comment your code, complete the code template format given.

Paste your complete MIPS code answer here

```
 # Title:       hello,world

# Author:      Qimin Zhu 6888143

# Date:        02/09/2026

# Description: greeting the user

# Input:       Anker

# Output:      Hello, Anker


################## Data segment ##################

.data

starter: .asciiz "Hello, "

prompt:  .asciiz "Enter your name: "         # Your strings / variables here

name:  .space 64

################## Code segment ##################

.text

.globl main                  # mark the “main” function as globally visible.



main:                        # main function entry

    li   $v0, 4              # Service 4 = print string
    la   $a0, prompt         # Load the address of the prompt text
    syscall                  # Print the prompt

    li   $v0, 8
    la   $a0, name
    li   $a1, 64
    syscall

    li   $v0, 4
    la   $a0, starter
    syscall

    li   $v0, 4              # Service 4 = print string
    la   $a0, name         # Load the address of the result text
    syscall                  # Print the result text
```

Question Q1.4

Question / Instruction

Q1.4
Write a MIPS program that executes the statement: ***s = (a + b) − (c + 101)***, where *a*, *b*, and *c* are user-provided integer inputs, and *s* is computed and printed as an output.

Comment your code, complete the code template format given.

Answer sub-questions (a)–(d) in the second box below.

Paste your complete MIPS code answer here

```
 # Title:       s = (a + b) – (c + 101)

# Author:      Qimin Zhu 6888143

# Date:        02/09/2026

# Description: calculation

# Input:       a, b, c

# Output:      s


################## Data segment ##################

.data

    prompt_a:  .asciiz "Enter the value of a: "         # Your strings / variables here
    prompt_b:  .asciiz "Enter the value of b: "
    prompt_c:  .asciiz "Enter the value of c: "
    result:  .asciiz "Result s = "

################## Code segment ##################

.text

.globl main                  # mark the “main” function as globally visible.



main:                        # main function entry

    li   $v0, 4
    la   $a0, prompt_a
    syscall

    li   $v0, 5
    syscall
    move $t0, $v0

    li   $v0, 4
    la   $a0, prompt_b
    syscall

    li   $v0, 5
    syscall
    move $t1, $v0

    li   $v0, 4
    la   $a0, prompt_c
    syscall

    li   $v0, 5
    syscall
    move $t2, $v0

    add $t3, $t0, $t1
    addi $t4, $t2, 101
    sub $t5, $t3, $t4

    li   $v0, 4
    la   $a0, result
    syscall

    li   $v0, 1
    move   $a0, $t5
```

```
    syscall

    li   $v0, 10
    syscall
```

Q1.4 Written answers:

a. Suppose the user enters a = 5, b = 10, and c = −30. What is the expected value of s?	*-56*
b. Which instruction in your program computed the value of s, and which register is used?	sub $t5, $t3, $t4
c. What is the address of this instruction in memory?	*0x0040005c*
d. Put a breakpoint at this instruction and write the value of the register used for computing s, in decimal and in hexadecimal.	*-56, 0xffffffc8*

Question Q1.5

Question / Instruction

Q1.5
Write a MIPS program that inputs two integer values.

The program should output **`equal`** if the two integers are equal. Otherwise, it should output **`not equal`**. Use the branch instruction to check for equality.

Comment your code, complete the code template format given.

```
Paste your complete MIPS code answer here
 # Title:       Euql


# Author:      Qimin Zhu 6888143


# Date:        02/09/2026


# Description: It checks if a and b equal


# Input:       2


# Output:      2



################## Data segment ##################


.data


prompt_a:  .asciiz "Enter an integer a: "         # Your strings / variables here

prompt_b:  .asciiz "Enter an integer b: "         # Your strings / variables here

is_equal:  .asciiz "They are equal.\n"
not_equal:  .asciiz "They are not euqal.\n"


################## Code segment ##################


.text


.globl main                  # mark the “main” function as globally visible.







main:                        # main function entry


    li   $v0, 4

    la   $a0, prompt_a

    syscall
```

```
    li   $v0, 5
    syscall
    move $t0, $v0

    li   $v0, 4
    la   $a0, prompt_b
    syscall

    li   $v0, 5
    syscall
    move $t1, $v0

    beq $t0, $t1, result

    li   $v0, 4
    la   $a0, not_equal
    syscall
    j   exit_program

result:

    li   $v0, 4
    la   $a0, is_equal
    syscall

exit_program:

    li   $v0, 10
    syscall
```

Congratulations!

You have written your first MIPS assembly programs: you used system calls for input and output, arithmetic instructions, branches, and the MARS debugger with breakpoints to inspect registers and memory.

Save this worksheet as PDF, add your Student ID into the filename and upload it to the Assignment page on MyCourses.